

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2024

EC2105-1 – NETWORK THEORY AND CONTROL SYSTEMS

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note: Usage of Graph sheet and Semilog sheet is permitted.

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Ideal current source has**
A) Zero internal resistance
B) Infinite internal resistance
C) Low value of voltage
D) Large value of current
2. **Mesh analysis is based on**
A) KVL
B) Both
C) KCL
D) Law of conservation of energy
3. **Which of the following is not a bilateral element?**
A) Constant current source
B) Resistor
C) Capacitor
D) Inductor
4. **Application of Norton's theorem to a circuit yield**
A) Equivalent current source and impedance in series
B) Equivalent current source and impedance in parallel
C) Equivalent impedance
D) Equivalent current source
5. **Superposition theorem can be applied only to circuits having**
A) Resistive elements
B) Passive elements
C) Nonlinear elements
D) Linear bilateral elements
6. **Under the conditions of maximum power transfer, the efficiency is _____.**
A) 75%
B) 50%
C) 100%
D) 25%
7. **Thevenin's theorem applies to**
A) one port network
B) two port network
C) three port network
D) for port network
8. **Ideal voltage source have**
A) Zero internal resistance
B) Infinite internal resistance
C) Low value of current
D) Large value of e.m.f
9. **The unit in which the Z-parameter are expressed is _____**
A) Mho
B) Farad
C) Ohm
D) Henry
10. **The hybrid parameter h_{11} is called?**
A) short circuit input impedance
B) open circuit reverse voltage gain
C) short circuit forward current gain
D) open circuit output admittance
11. **The net current through a port is**
A) Infinity
B) Unity
C) Zero
D) None of the above

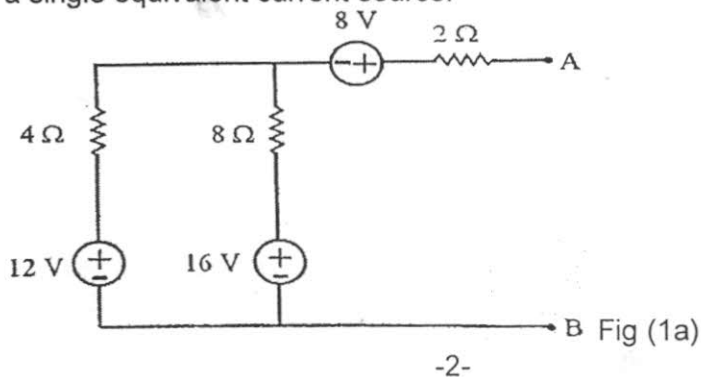
12. Routh Hurwitz criterion gives
- A) Number of roots in the right half of the s-plane
B) Value of the roots
C) Number of roots in the left half of the s-plane
D) Number of roots in the top half of the s-plane
13. Loop which does not possess any common node are said to be _____ loops
- A) Forward gain
B) Touching loops
C) Non touching loops
D) Feedback gain
14. Which one of the following statements is not correct?
- A) Root loci can be used for analyzing stability and transient performance
B) Shape of the root locus gives idea of type of controller needed to meet design specification
C) Root loci provide insight into system stability and performance
D) Root locus can be used to handle more than one variable at a time
15. What is the number of the root locus segments which do not terminate on zeroes?
- A) The sum of the number of poles and the number of the zeroes
B) The number of poles
C) The number of zeroes
D) The difference between the number of poles and zeroes
16. When the number of poles is equal to the number of zeroes, how many branches of root locus tends towards infinity?
- A) 0
B) 1
C) 2
D) None of the above
17. Cut off frequency is the frequency at which magnitude of closed loop frequency response is
- A) 1 dB below its zero frequency
B) 2 dB below its zero frequency
C) 3 dB below its zero frequency
D) 4 dB below its zero frequency
18. For relative stability of the system which of the following is sufficient?
- A) Gain Margin
B) Phase Margin
C) Both (a) and (b)
D) None of the above
19. Gain margin of a first or second order system is
- A) 0
B) ∞
C) 1
D) 10
20. In the Bode plot, the frequency where the two asymptotic lines meet is called
- A) Linear frequency
B) Phase cross over frequency
C) Gain cross over frequency
D) Corner frequency

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Using source transformation reduce the circuit shown in Fig (1a) into a single equivalent current source.



8 L*3 1 1

- b) For the network shown in Fig (1b), find the current through 4Ω and 6Ω resistors using mesh analysis.

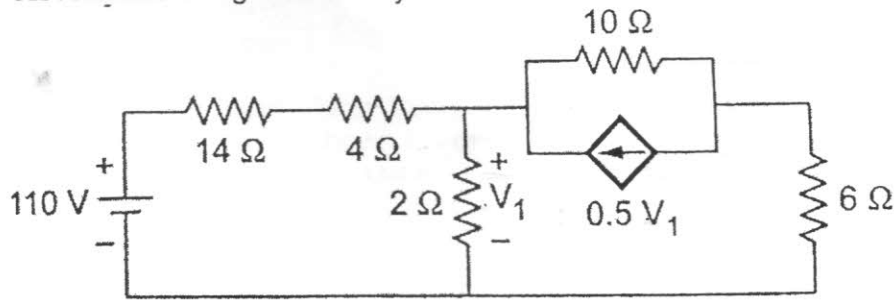


Fig (1b)

8 L3 1 1

2. a) Find the power delivered by the dependent voltage source for the network shown in Fig (2a).

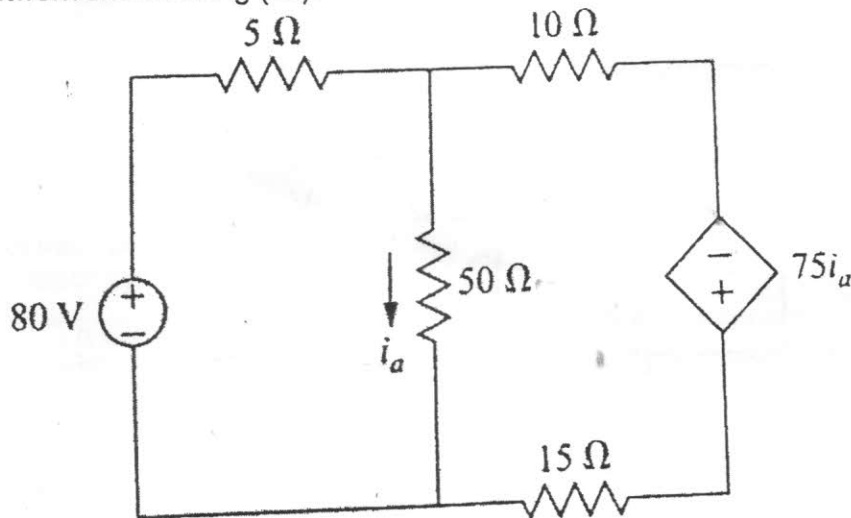


Fig (2a)

8 L3 1 1

- b) Find the mesh currents for the network shown in Fig (2b).

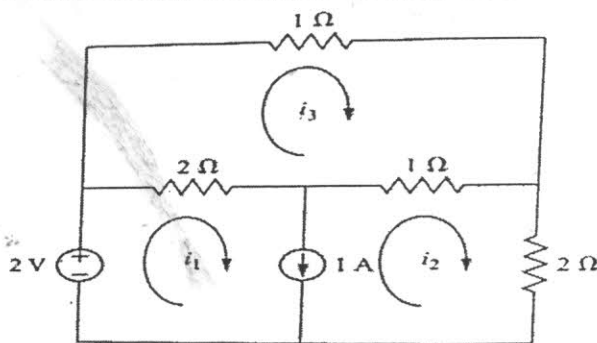


Fig (2b)

8 L3 1 1

3. a) For the network shown in Fig (3a), solve for v_x using Superposition theorem.

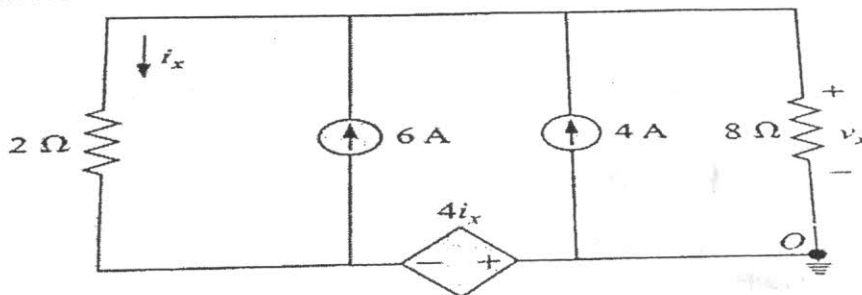


Fig (3a)

8 L3 2 1

- b) Find I_0 for the network shown in Fig (3b) using Norton's theorem.

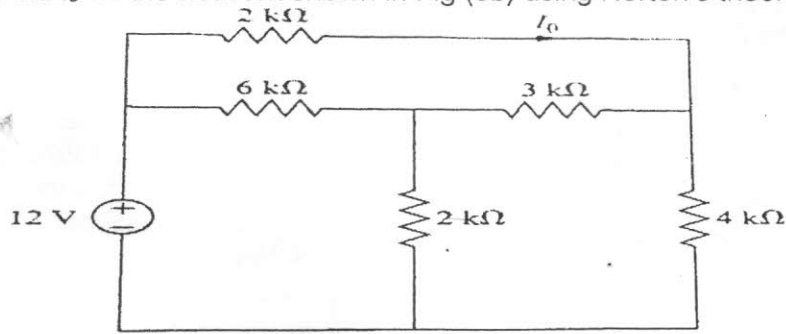


Fig (3b)

8 L3 2 1

Unit – II

4. a) Find Z parameters for the network shown in Fig (4a).

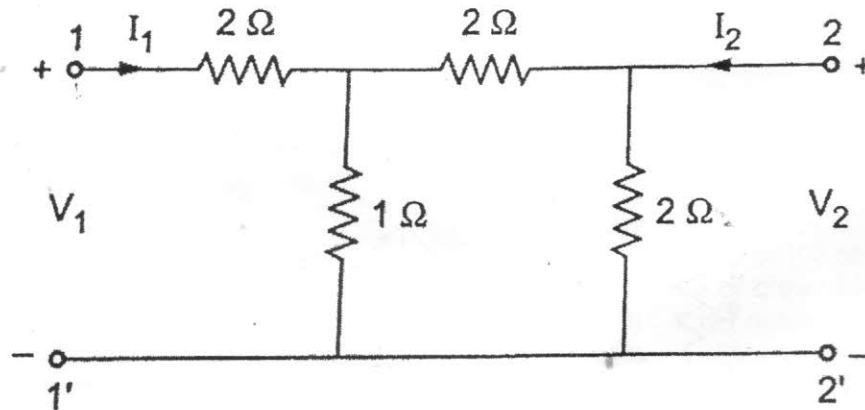


Fig (4a)

8 L3 3 1

- b) Express h-parameters in terms of ABCD parameters.

8 L2 3 1

5. a) Find $\frac{C}{R}$ for the graph shown in Fig (5a) using Mason's gain Formula.

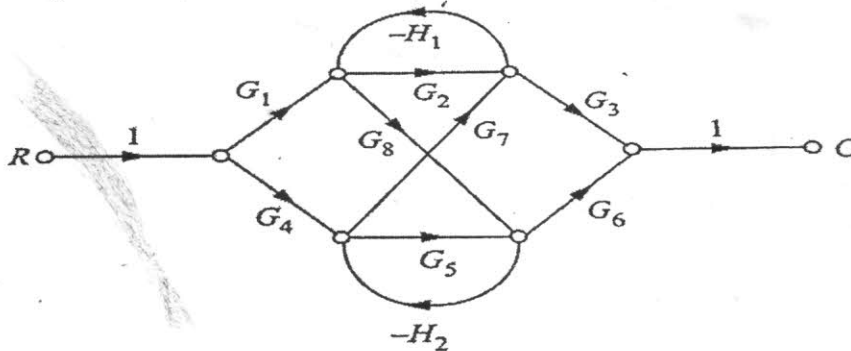


Fig (5a)

8 L3 4 1

- b) Discuss the advantages and disadvantages of open loop control system and closed loop control system.

8 L1 4 1

6. a) The open loop transfer function of a unity feedback system is given by, $G(s) = \frac{10(s+1)}{s(s-1)(s+5)}$. Determine the stability of the closed loop system using Routh Hurwitz criterion.

6 L3 4 1

- b) Draw the Root Locus and determine the stability of the system with

$$GH(s) = \frac{K}{s(s+2)(s+3)}$$

10 L3 4 1

Unit – III

7. a) Construct Bode magnitude and phase diagrams for $GH(s) = \frac{100(0.1s+1)}{s(s+1)^2(0.01s+1)}$. Comment on the closed-loop stability of the system.
- b) Determine the open loop transfer function of a system whose approximate plot is as shown in Fig (7b).

10 L3 5 1

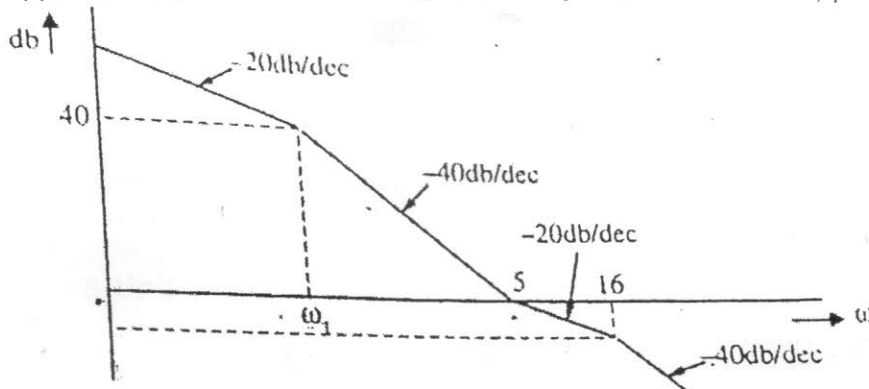


Fig (7b)

6 L3 5 1

8. a) Plot the Bode diagram for the open-loop transfer function of a unity feedback system given below, $G(s) = \frac{4}{(0.1s+1)^2(0.01s+1)}$. Comment on the stability of the system.
- b) Determine the open loop transfer function of a system whose approximate plot is as shown in Fig (8b).

10 L3 5 1

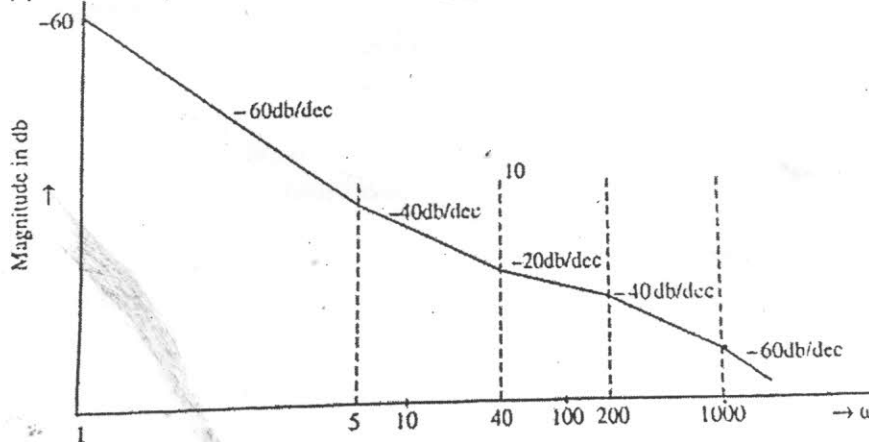


Fig (8b)

6 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
